DG INTERNS HUB – CYBERSECURITY INTERNSHIP

WEEK 4 ASSIGNMENT REPORT:

UNDERSTAND BASIC NETWORK SETUP AND PERFORM INITIAL SECUROTY TESTING IN A LAB ENVIRONMENT.

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Sr. No.	Particulars	Brief Description
1.	Task 1 – Setup Basic Network (Lab)	Setup of the cybersecurity lab using Kali Linux and Ubuntu in VirtualBox, including Host-Only networking, IP configuration, and connectivity testing.
2.	Task 2 – Install and Configure Services	Installation and configuration of the Apache web server on Ubuntu, followed by service verification and access testing from Kali Linux.
3.	Task 3 – Network Scanning Using Nmap	Network scanning using Nmap to identify active hosts, open ports, and running services, along with saving and analyzing scan results.
4.	Task 4 – Traffic Analysis Using Wireshark	Capture and analysis of network traffic using Wireshark, focusing on ICMP, HTTP, and DNS protocols.
5.	Task 5 – Mini Project Output	Integration of the completed activities into a practical cybersecurity workflow covering network setup, service configuration, scanning, and traffic analysis.
6.	Overall Key Findings	Summary of the major technical findings obtained from all five tasks.
7.	Overall Learning Outcomes	Summary of the networking and cybersecurity knowledge and practical skills gained during the activities.
8.	Overall Conclusion	Overall summary of the Week 4 activities and the practical cybersecurity experience gained from the complete lab workflow.

Task 1: Setup Basic Network (Lab)

A controlled network laboratory was established using Oracle VirtualBox to understand the basic configuration and communication of systems within a virtual network environment.
The laboratory consisted of two virtual machines: Kali Linux, configured as the attacker/security testing machine, and Ubuntu Linux, configured as the target machine. Both systems were connected through a VirtualBox Host-only network, allowing them to communicate with each other within an isolated virtual environment.
The configured environment provides a safe and controlled platform for performing authorized cybersecurity and network-security experiments.

Objective

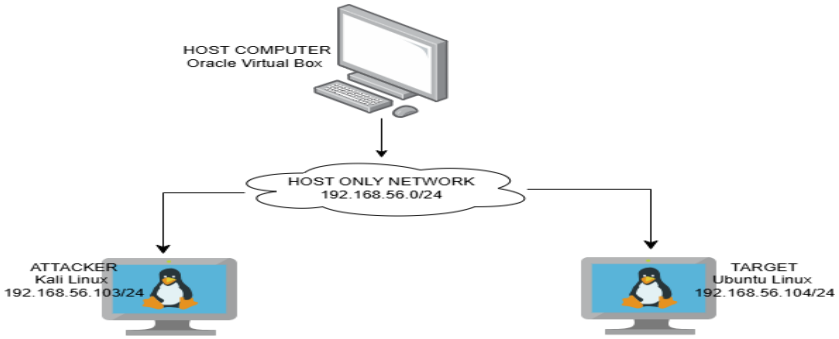
- The objectives of this task were:
- To create a basic virtual network using Oracle VirtualBox.
- To configure Kali Linux as the attacker/security testing machine.
- To configure Ubuntu Linux as the target machine.
- To connect both virtual machines to the same virtual network.
- To identify the IP address and subnet configuration of each machine.
- To verify communication between the two systems.
- To perform basic host discovery using Nmap.
- To establish a controlled environment for subsequent cybersecurity laboratory activities.

Lab Environment

Component	Configuration
Virtualization Platform	Oracle VirtualBox
Attacker Machine	Kali Linux
Target Machine	Ubuntu Linux
Number of Virtual Machines	2
Network Type	Host-only Adapter
Virtual Network	VitualBox Host only Ethernet adapter
Network Address	192.168.56.1/24
Network Testing Tool	Nmap
Connectivity Testing	ICMP Ping

Network Topology

The laboratory was designed using a Host-only virtual network. Kali Linux and Ubuntu Linux were connected to the same virtual network and assigned IP addresses within the same subnet.

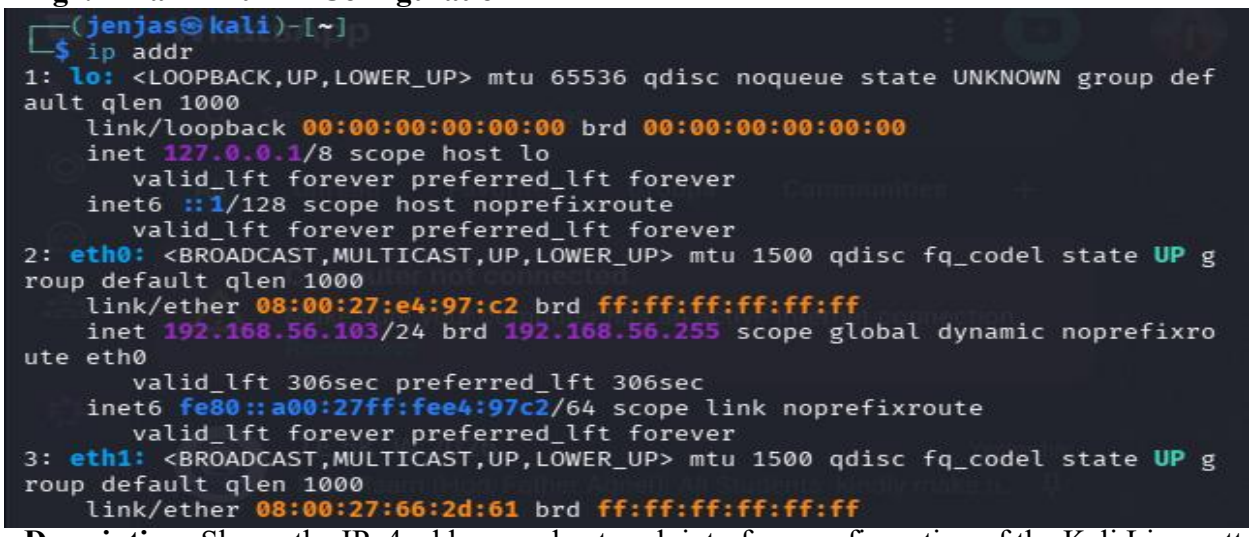


Network Configuration

Kali Linux – Attacker Machine

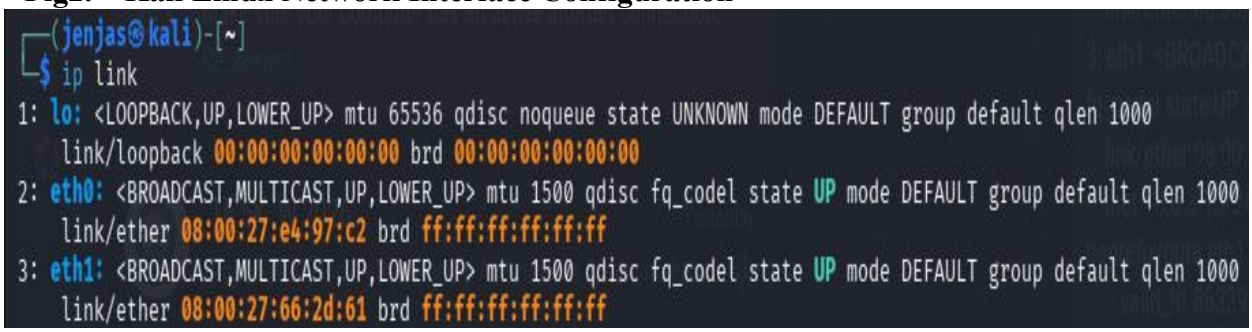
Kali Linux was configured as the attacker/security testing machine.
The VirtualBox network adapter was configured using the Host-only Adapter option. The adapter was connected to the VitualBox Host only Ethernet adapter virtual network.

Fig1. – Kali Linux IP Configuration



Description: Shows the IPv4 address and network interface configuration of the Kali Linux attacker machine.
The IP configuration was verified using: ip -4 addr

Fig2. – Kali Linux Network Interface Configuration



Description: The output shows the active network interfaces of Kali Linux, including eth0 configured for the Host-only network and eth1 configured for the NAT connection.
The network interface was also checked using: ip link

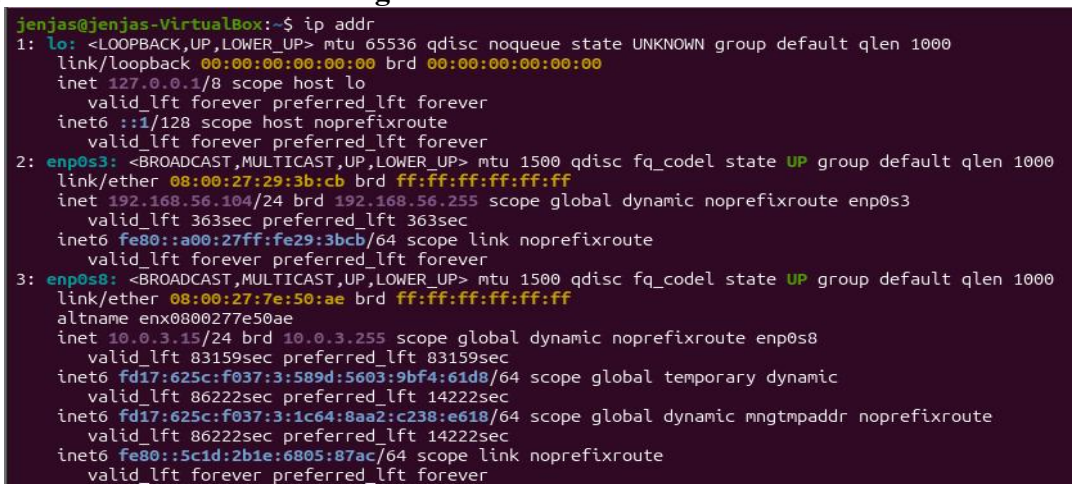
Kali Linux Configuration

Parameter	Value
Machine Role	**Attacker**
Operating System	**Kali Linux**
Network Adapter	**Host-only Adapter**
Interface	**Eth0**
IPv4 Address	**192.168.56.103/24**
Subnet Mask	**255.255.255.0**
Network	**192.168.56.0/24**

Ubuntu Linux – Target Machine

Ubuntu Linux was configured as the target machine for the controlled cybersecurity laboratory.
The VirtualBox network adapter was configured with the same Host-only Adapter network used by Kali Linux.

Fig3. – Ubuntu Linux IP Configuration



Description: Shows the IPv4 address and network interface configuration of the Ubuntu Linux attacker machine.
The IP configuration was verified using: ip -4 addr

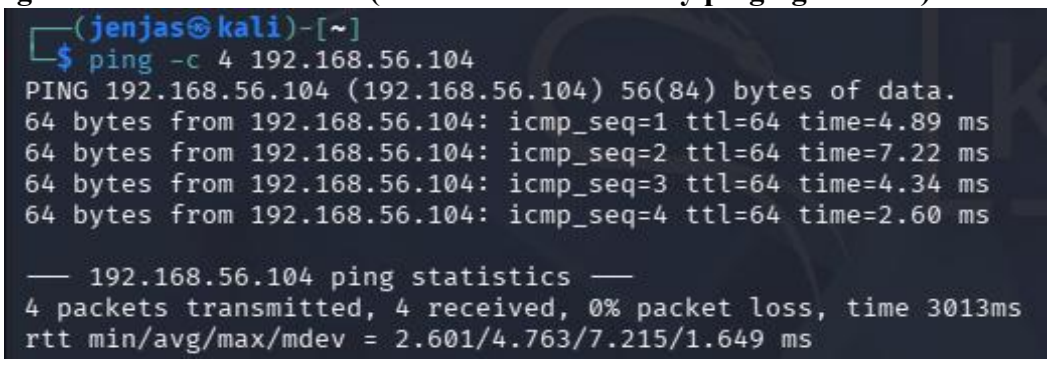
Ubuntu Configuration

Parameter	Value
Machine Role	**Target**
Operating System	**Ubuntu Linux**
Network Adapter	**Host-only Adapter**
Interface	**Enp0s3**
IPv4 Address	**192.168.56.104/24**
Subnet Mask	**255.255.255.0**
Network	**192.168.56.0/24**

Connectivity Verification

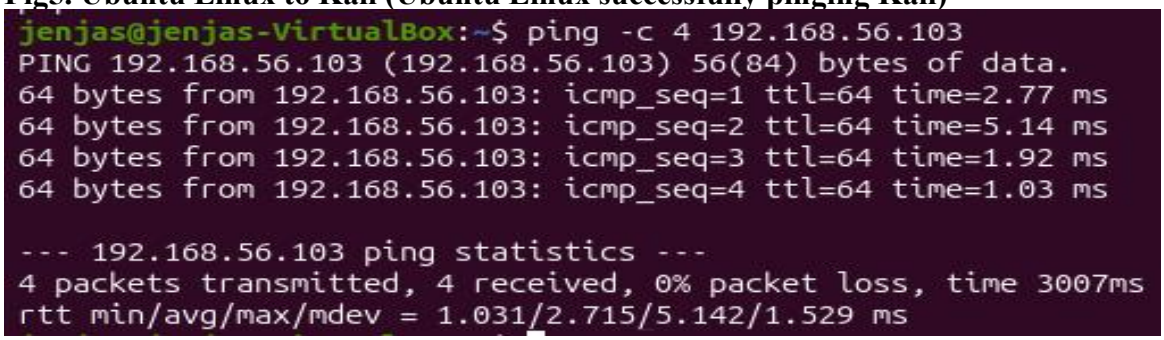
After configuring both virtual machines, connectivity between Kali Linux and Ubuntu Linux was tested using the ICMP protocol.

Fig4. Kali Linux to Ubuntu (Kali Linux successfully pinging Ubuntu)



Command: ping -c 4 192.168.56.104
Result: The Ubuntu machine responded successfully to the ICMP requests.

Fig5. Ubuntu Linux to Kali (Ubuntu Linux successfully pinging Kali)

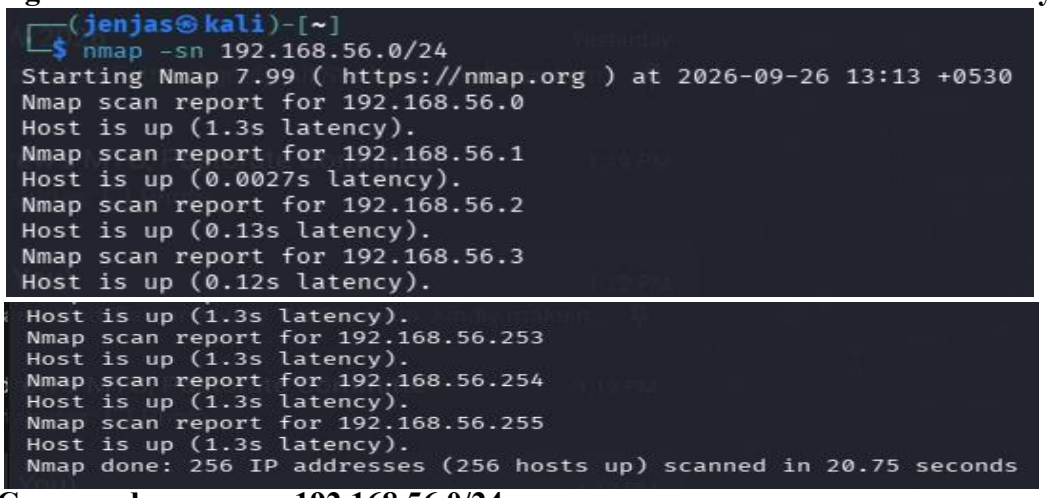


Command: ping -c 4 192.168.56.104
Result: The Kali Linux machine successfully responded to the ICMP requests.

Observation: Packets were successfully transmitted and received, confirming basic network connectivity.
This confirmed bidirectional communication between the two virtual machines.

Host Discovery Using Nmap
After confirming connectivity, Nmap was used from the Kali Linux machine to identify active hosts within the authorized laboratory network.

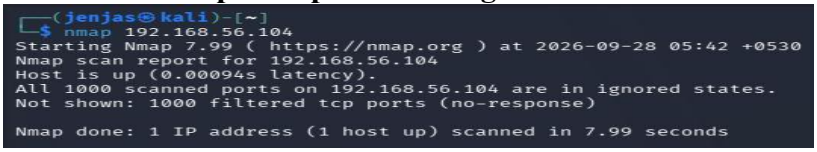
Fig6. Shows the identification of active hosts within the authorized laboratory subnet.



Command: nmap -sn 192.168.56.0/24
Result: Nmap successfully identified active systems within the configured Host-only network.
The Ubuntu virtual machine was identified as the target system within the authorized laboratory environment.

Basic Target Scan
After identifying the Ubuntu target, a basic Nmap scan was performed against its IP address:

Fig7. Shows the basic Nmap scan performed against the authorized Ubuntu target.



Command used: nmap 192.168.56.104
The scan was performed only against the Ubuntu virtual machine configured as part of the authorized laboratory.

Observation

The scan provided information about the network services exposed by the Ubuntu target within the laboratory environment. This information can be used in later cybersecurity tasks to understand network attack surfaces and service exposure.

Key Findings

The following findings were obtained during the laboratory exercise:

1. A functional virtual network was successfully created using Oracle VirtualBox.
2. Kali Linux and Ubuntu Linux were successfully connected to the same Host-only network.
3. Both virtual machines were assigned IPv4 addresses within the same subnet.
4. Successful ICMP communication was established between the two machines.
5. Nmap successfully identified active hosts within the laboratory subnet.
6. A basic scan of the authorized Ubuntu target provided information about its exposed network services.
7. The isolated environment provides a suitable foundation for further authorized cybersecurity and network-security experiments.

Security Considerations

The network laboratory was configured using a Host-only network to provide a controlled environment for cybersecurity testing.
All network discovery and scanning activities were performed only against virtual machines created and controlled for the laboratory exercise.
The following security practices were followed:

- Testing was restricted to the authorized virtual laboratory.
- The target system was a dedicated Ubuntu virtual machine.
- Network discovery was performed only against the configured laboratory subnet.
- No unauthorized external systems were scanned.
- The Host-only configuration helped isolate the laboratory traffic from the external network.

Learning Outcomes

Through this task, the following concepts were understood and practiced:

- Virtual network configuration
- Host-only networking
- IPv4 addressing
- Subnet identification
- Network interface configuration
- ICMP connectivity testing
- Basic network discovery
- Nmap host discovery
- Basic service enumeration
- Attacker and target laboratory architecture
- Safe and authorized cybersecurity testing

Conclusion

The basic network laboratory was successfully established using Oracle VirtualBox, with Kali Linux configured as the attacker and Ubuntu Linux as the target. Both systems were connected through a Host-only network, and connectivity was verified using ICMP and Nmap. The setup provides a controlled environment for further authorized cybersecurity testing.

Task 2 – Install and Configure Services

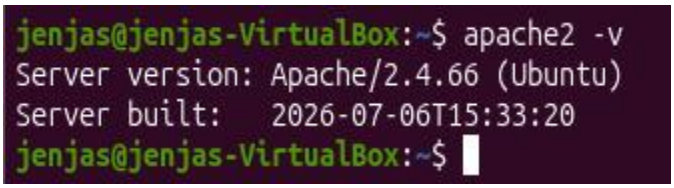
To install a basic web server on the Ubuntu target machine, verify that the service is running correctly, and test its accessibility from Kali Linux.

Apache Web Server Installation

Apache2 was selected as the basic web server for the laboratory.
On Ubuntu Commands used:
sudo apt update:
sudo apt install apache2 -y : Apache was then installed using

Note about the installation
During installation, an APT dependency issue was encountered. The package manager reported unmet dependencies. The package system was repaired using: **sudo apt --fix-broken install**
After repairing the dependencies,
The Apache installation was then attempted again: **sudo apt install apache2 -y**

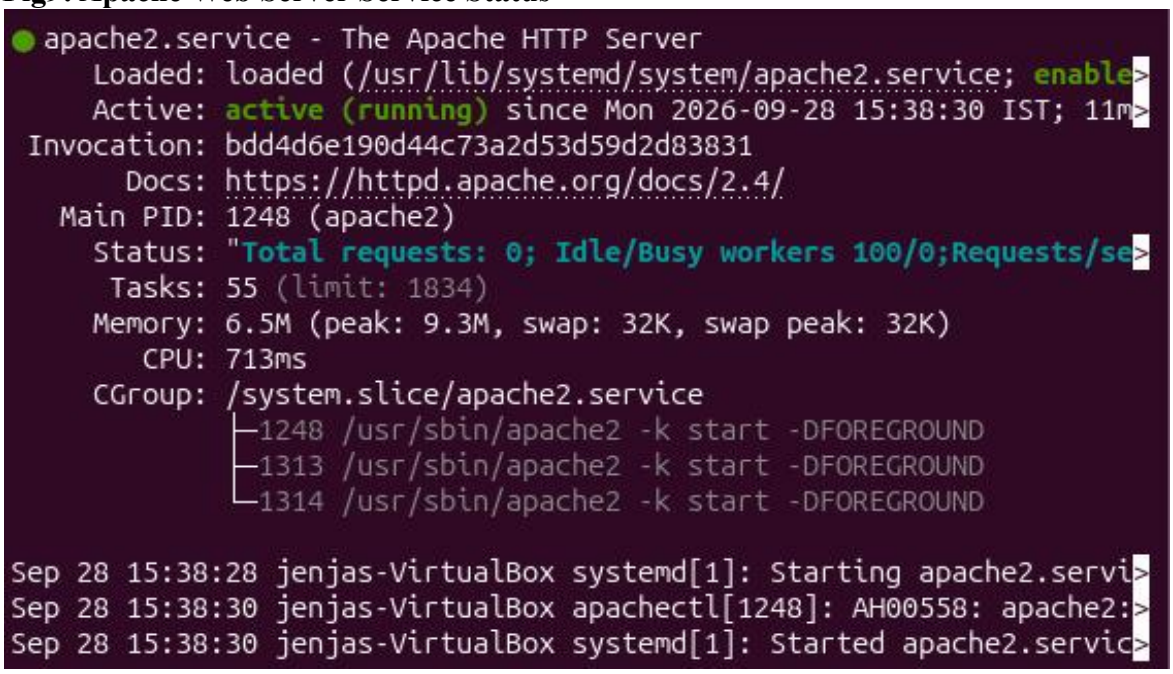
Fig8: Apache Web Server Installation on Ubuntu



Apache Service Verification

After installation, the Apache service was checked using: **sudo systemctl status apache2**
The expected status is: **Active: active (running)**

Fig9: Apache Web Server Service Status

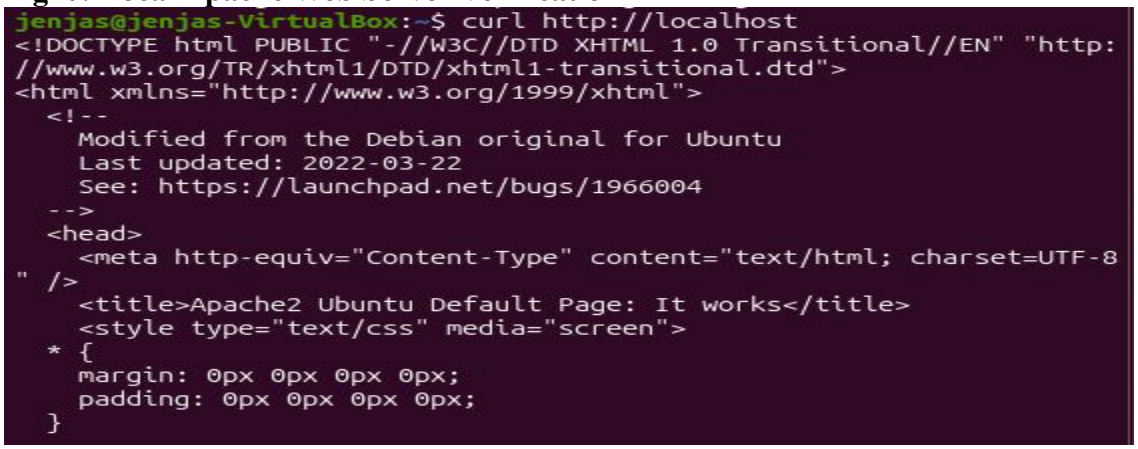


This confirms that the Apache web server is running.

Local Web Server Verification

The Apache server was tested locally on Ubuntu using: **curl http://localhost**
The command returned HTML content from the Apache default webpage, confirming that the web server was responding locally.
Alternatively, the following could be entered in the Ubuntu browser: **http://localhost**

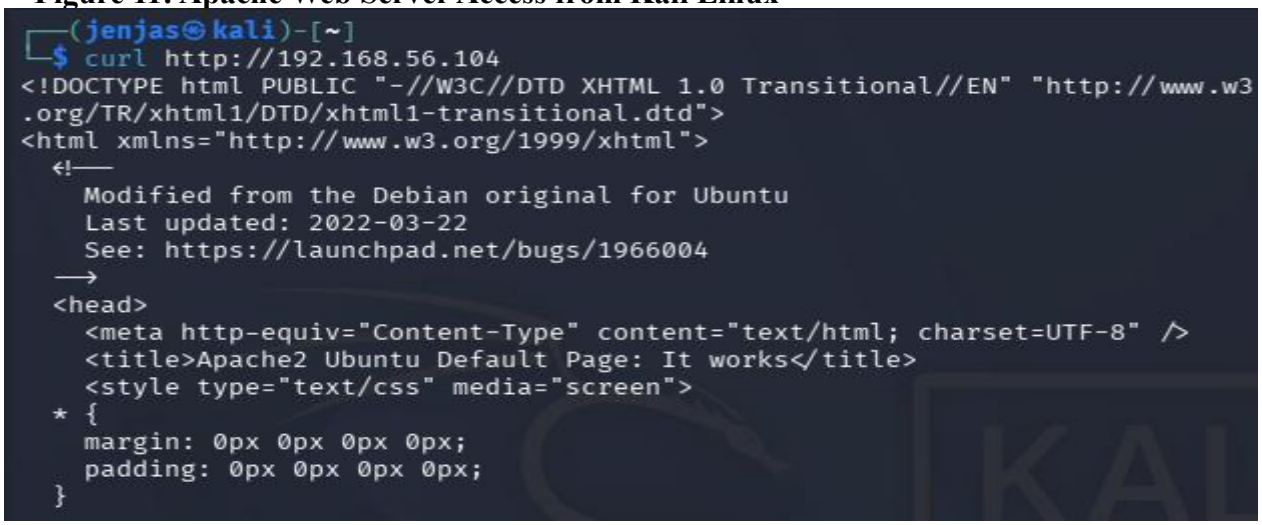
Fig10: Local Apache Web Server Verification



Remote Web Server Verification from Kali

The Ubuntu Host-only IP was used to access the Apache server from Kali.
Command: curl http://192.168.56.104
If Apache is working correctly, HTML content from the Ubuntu web server should be displayed.

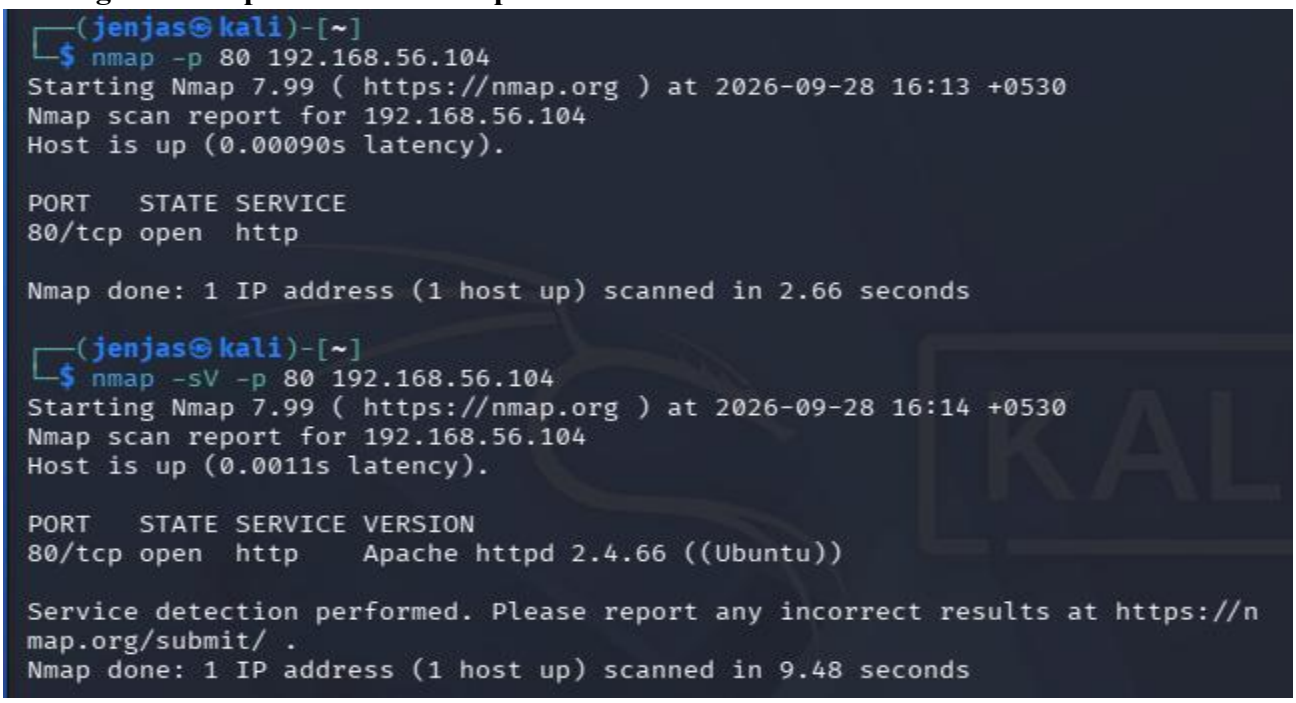
Figure 11: Apache Web Server Access from Kali Linux



Nmap Verification of HTTP Service

Nmap was used to verify whether the HTTP service was accessible on the Ubuntu target.
From Kali:
nmap -p 80 192.168.56.204
A service/version scan can also be performed:
nmap -sV -p 80 192.168.56.104

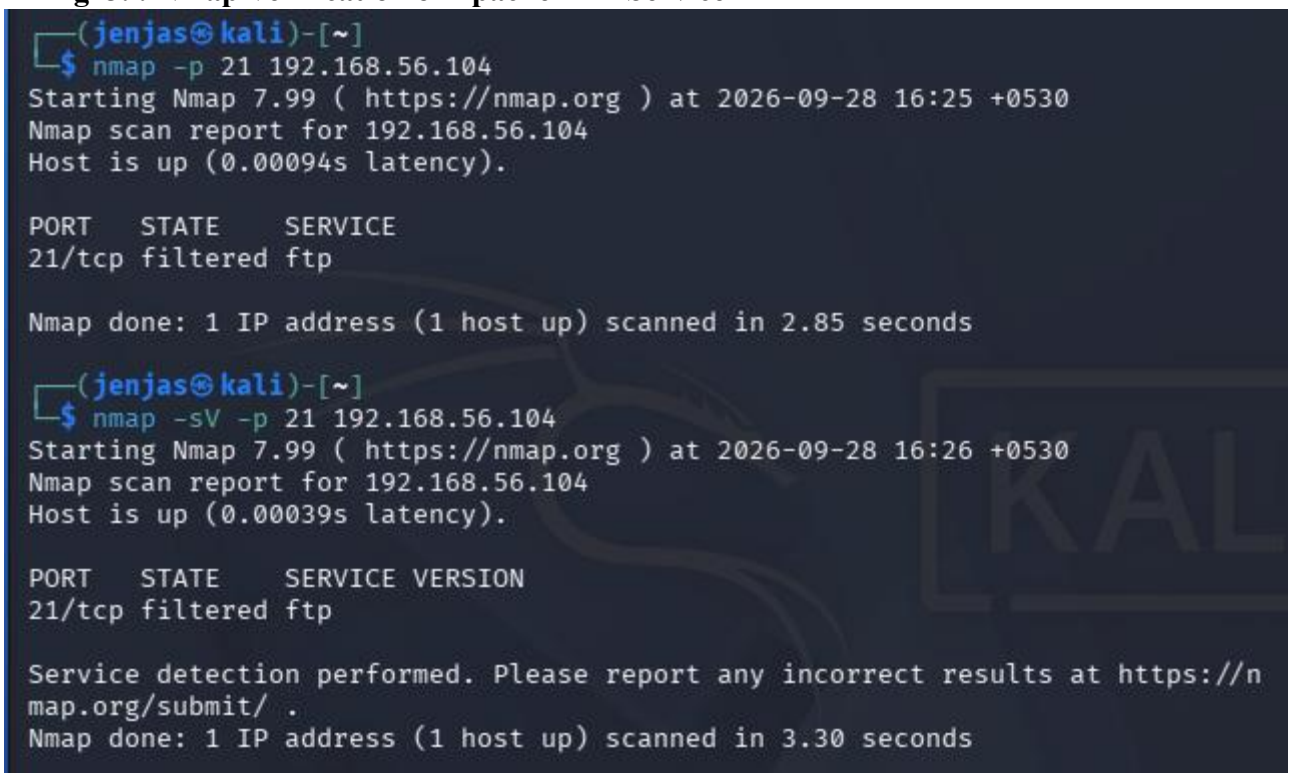
Fig12. : Nmap Verification of Apache HTTP Service



Optional Task – FTP Server

The FTP server was an optional component of the assessment.
The FTP service can then be checked from Kali using:
Command used: nmap -p 21 192.168.56.104 & nmap -sV -p 192.168.56.104

Fig13. : Nmap Verification of Apache FTP Service



Key Findings

- A basic virtual network was successfully configured using Oracle VirtualBox.
- Kali Linux was configured as the attacker machine and Ubuntu Linux as the target machine.
- Both systems were connected through the Host-only network 192.168.56.0/24.
- Connectivity between the systems was verified using ICMP ping.
- Nmap was successfully used for host discovery and basic target scanning.
- Apache2 was configured on Ubuntu as a basic web service and verified using service and network tests.

Security Considerations

All network scanning and service-testing activities were performed within the controlled virtual laboratory environment. The Host-only network helped isolate the testing traffic, and Nmap scans were limited to authorized virtual machines. No unauthorized external systems were targeted.

Observations

The virtual machines were able to communicate successfully through the Host-only network. Nmap identified active hosts and provided information about the target's accessible services. The Apache web server on Ubuntu was observed to respond to HTTP requests, demonstrating successful service configuration and network accessibility.

Conclusion

The Week 4 assessment successfully established a controlled cybersecurity laboratory using Kali Linux and Ubuntu Linux. Network connectivity, host discovery, target scanning, and basic web-service configuration were successfully practiced, providing a foundation for further cybersecurity exercises.

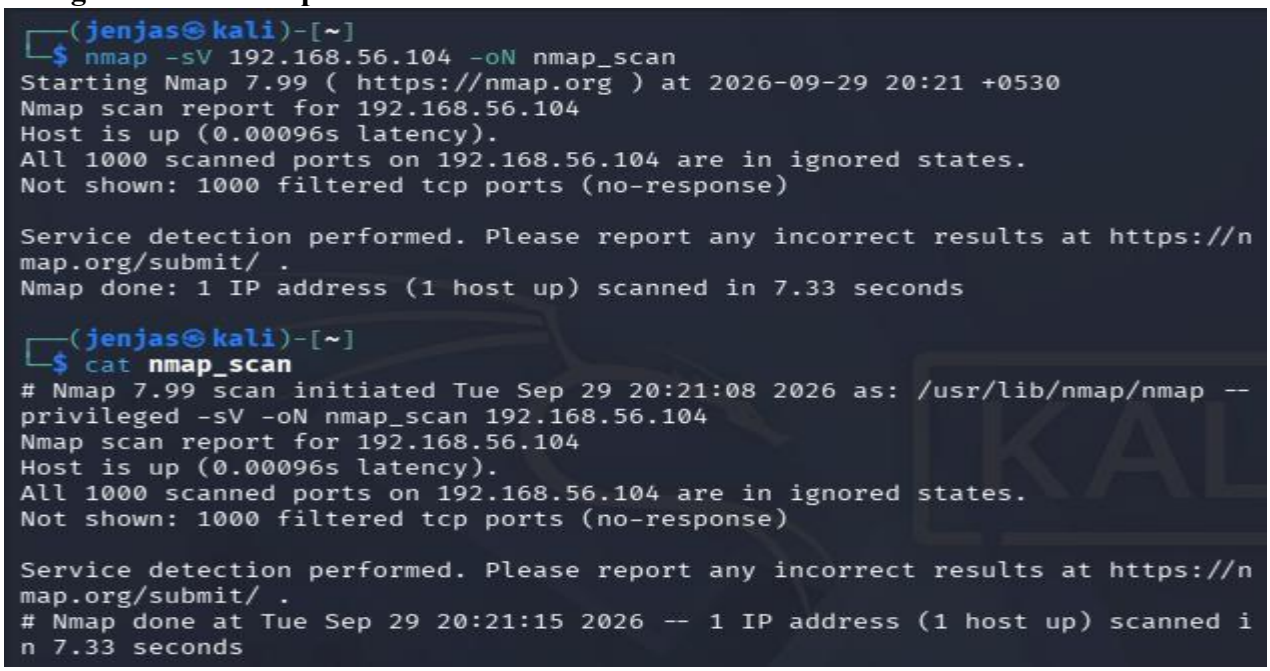
Task 3: Network Scanning Using Nmap

To perform network scanning of the Ubuntu target using Nmap, identify open ports and running services, detect service versions, and save the scan results for analysis.

Some of the subtasks under **Task 3** were already completed as part of the previous tasks.

Saving Scan Results

Fig.15: Saved Nmap Scan Results



Key Findings

- Nmap successfully scanned the Ubuntu target.
- Open and closed/filtered ports were identified.
- Running services were identified using service detection.
- The Apache HTTP service was verified on port 80 when accessible.
- The scan results were saved for documentation and analysis.

Observation

The Nmap scan provided information about the services exposed by the Ubuntu target. Service-version detection gave additional details about the applications associated with the detected open ports.

Security Consideration

The Nmap scan was performed only against the authorized Ubuntu machine within the controlled Host-only laboratory network. Identifying unnecessary open ports and services can help administrators reduce the system's exposed attack surface.

Conclusion

The Ubuntu target was successfully scanned using Nmap from Kali Linux. Open ports and running services were identified, service detection was performed, and the results were saved for further analysis.

Task 4: Traffic Analysis Using Wireshark

To capture and analyze network traffic using Wireshark and identify basic network protocols such as ICMP, HTTP, and DNS through appropriate packet-capture and display-filter techniques.

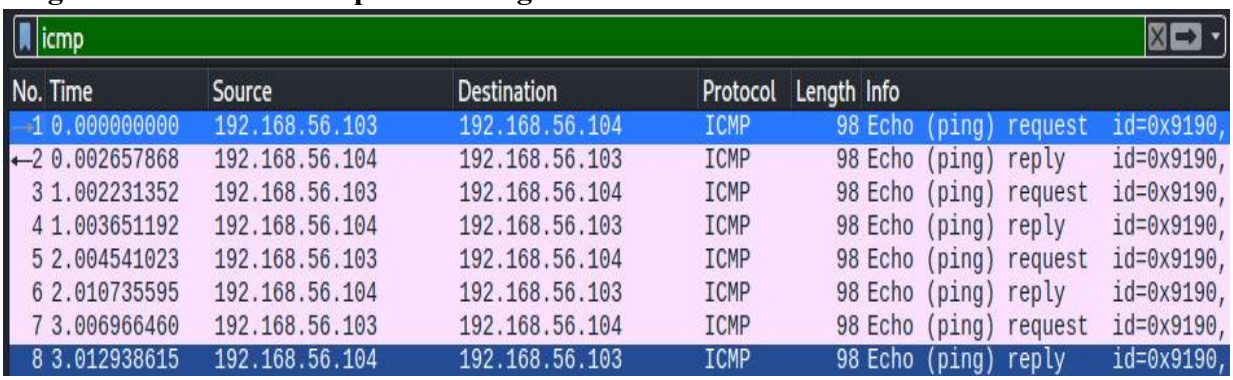
Network Traffic Capture

Wireshark was used on the Kali Linux virtual machine to capture network packets generated during communication with the Ubuntu target. The appropriate network interface was selected, and packet capture was initiated before generating network traffic.
The captured packets were subsequently analyzed using Wireshark display filters to isolate specific protocols.

ICMP Traffic Analysis

ICMP traffic was generated by performing a connectivity test between the Kali Linux and Ubuntu Linux virtual machines using the ping command.
Command Used
ping -c 4 <Ubuntu-IP>
The following Wireshark display filter was applied:
icmp
The captured packets showed ICMP Echo Request and Echo Reply messages exchanged between the two virtual machines. These packets demonstrate the successful exchange of ICMP traffic during the connectivity test.

Fig.16: ICMP Traffic Captured Using Wireshark

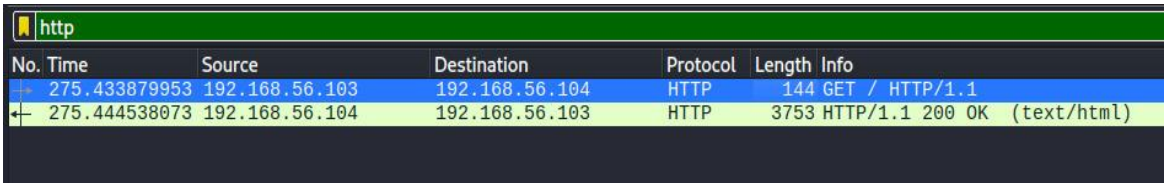


Observation:
The captured ICMP packets demonstrated the exchange of Echo Request and Echo Reply messages between the Kali Linux and Ubuntu Linux systems.

HTTP Traffic Analysis

HTTP traffic was generated by accessing the Apache web server running on the Ubuntu target from Kali Linux.
Command Used
curl http://<Ubuntu-IP>
The following Wireshark display filter was applied:
http
The captured traffic showed HTTP communication between the Kali Linux client and the Apache web server running on the Ubuntu target. The packets provided information about HTTP requests and responses exchanged during the communication.

Fig.17: HTTP Traffic Captured Using Wireshark

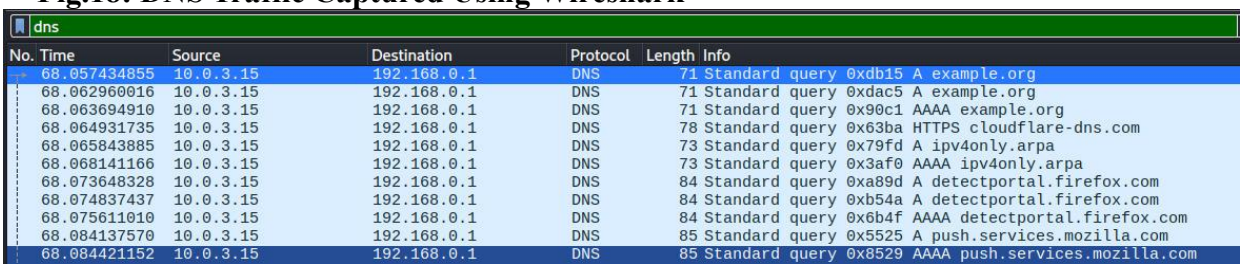


Observation:
The captured HTTP packets demonstrated the communication between the client and the Apache web server. The packet details provided information about the HTTP request and response exchange.

DNS Traffic Analysis

DNS traffic was generated by performing a domain name resolution request from Kali Linux.
Command Used
nslookup example.com
The following Wireshark display filter was applied:
dns
The captured packets showed DNS query and response traffic associated with domain name resolution. The packet details provided information about the DNS communication between the system and the configured DNS server.

Fig.18: DNS Traffic Captured Using Wireshark



Observation:
The captured DNS packets demonstrated the exchange of DNS queries and responses used to resolve a domain name into its corresponding IP address.

Key Findings

The following findings were obtained during the traffic analysis activity:

1. Network traffic was successfully captured using Wireshark.
2. ICMP packets were captured during the connectivity test between Kali Linux and Ubuntu Linux.
3. HTTP packets were captured while accessing the Apache web server hosted on the Ubuntu target.
4. DNS query and response packets were captured during domain name resolution.
5. Wireshark display filters were used to isolate and analyze specific network protocols.
6. The packet details provided practical information about source, destination, protocol, and communication type.

Conclusion
The Traffic Analysis task was successfully completed using Wireshark. Network traffic generated through ICMP, HTTP, and DNS activities was captured and analyzed using appropriate display filters. The exercise provided practical experience in packet capture, protocol identification, and basic network traffic analysis within a controlled cybersecurity laboratory environment.

Task 5: Mini Project Output

Objective

To demonstrate the complete workflow of the cybersecurity laboratory by integrating the network setup, configured services, Nmap scanning, and Wireshark traffic analysis performed in the previous tasks.

Mini Project Overview

The mini project demonstrates a basic cybersecurity assessment workflow using a controlled virtual network environment. Kali Linux was configured as the analysis and security testing machine, while Ubuntu Linux was configured as the target system.
The workflow consisted of establishing network connectivity, configuring a web service on the target, scanning the target using Nmap, and capturing and analyzing network traffic using Wireshark.
The complete workflow is represented as:
Network Setup → Service Configuration → Network Scanning → Traffic Capture and Analysis

Network Setup

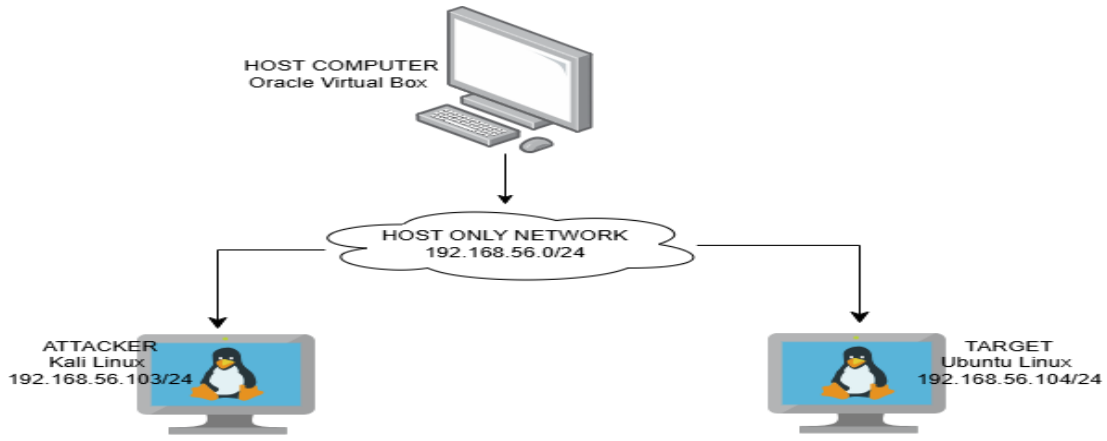
A controlled virtual network was established using Oracle VirtualBox. Kali Linux and Ubuntu Linux were connected through the Host-only network vboxnet0 using the 192.168.56.0/24 network.
Kali Linux was used as the attacker and analysis machine, while Ubuntu Linux was used as the target machine.

Network Configuration

Component	Configuration
Virtualization Platform	Oracle VirtualBox
Attacker / Analysis Machine	Kali Linux
Target Machine	Ubuntu Linux
Network Type	Host-only
Network Adapter	vboxnet0
Network Address	192.168.56.0/24
Kali Linux IP	192.168.56.101
Ubuntu Linux IP	<Ubuntu-IP>

Connectivity between the virtual machines was verified using ICMP traffic.

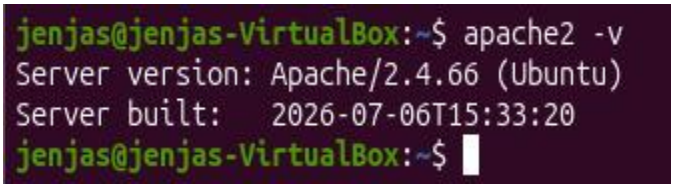
Fig. Complete Virtual Network Setup



Services Running

The Apache2 web server was installed and configured on the Ubuntu target machine. The service was started and verified to ensure that the web server was accessible from the laboratory network.
The Apache service was tested locally on Ubuntu and remotely from Kali Linux using the target's Host-only IP address.

Fig: Apache Web Server Running on Ubuntu



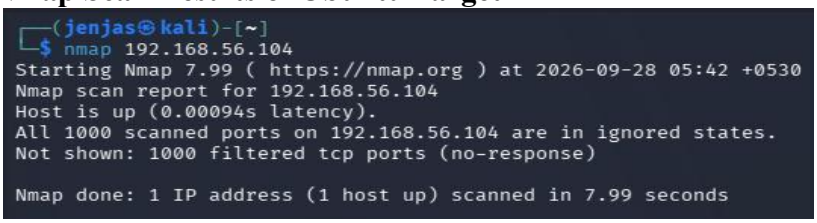
Figu: Apache Web Server Accessed from Kali Linux



Network Scanning

After configuring the target service, Nmap was used from Kali Linux to examine the Ubuntu target.
Basic scanning and service-version detection were performed to identify accessible ports and the services associated with them. The HTTP service provided by Apache was also verified through Nmap.
The scanning activity demonstrated how an authorized security tester can identify network services exposed by a target system.

Fig: Nmap Scan Results of Ubuntu Target



Traffic Capture and Analysis

Following the network scanning and service configuration activities, Wireshark was used to capture and analyze network traffic.
Three basic protocols were analyzed:

Protocol	Activity Used to Generate Traffic	Wireshark Filter
ICMP	ping -c 4 <Ubuntu-IP>	icmp
HTTP	curl http://<Ubuntu-IP>	http
DNS	nslookup example.com	dns

The captured packets were examined to understand the communication between systems and the characteristics of different network protocols.

Fig: ICMP Traffic Captured Using Wireshark

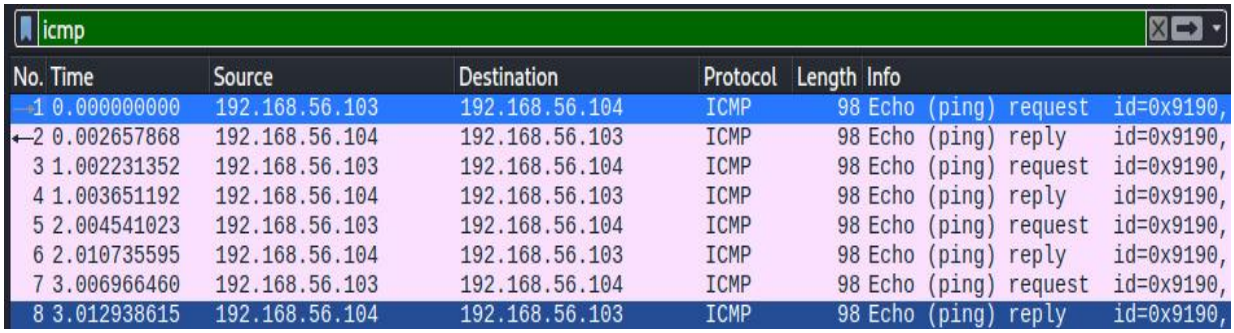


Fig: HTTP Traffic Captured Using Wireshark

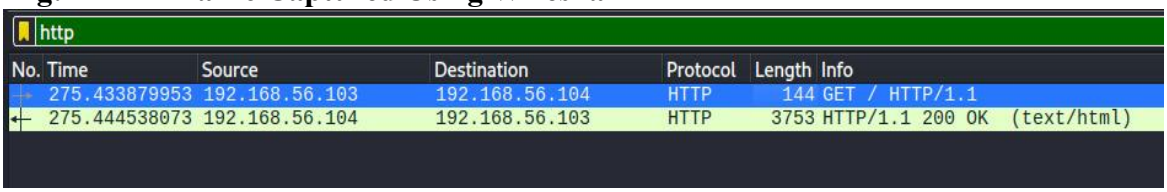
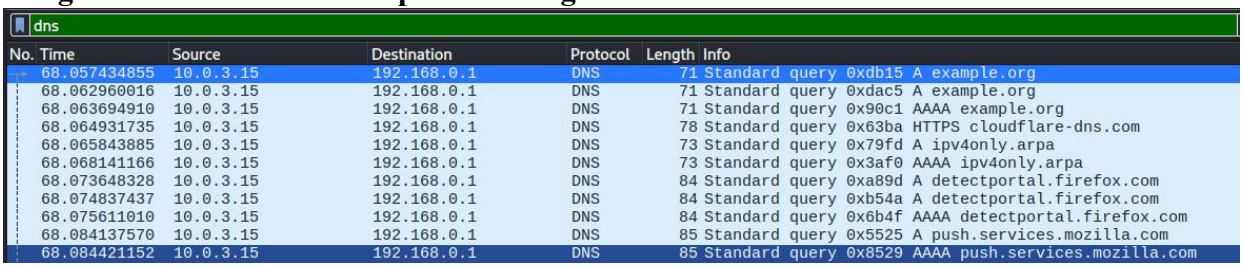


Figure 5.7: DNS Traffic Captured Using Wireshark



Complete Workflow

The overall workflow of the mini project can be summarized as follows:

Stage	Activity	Tool / Technology	Output
1	Network setup	Oracle VirtualBox	Kali and Ubuntu connected through Host-only network
2	Connectivity verification	ping	Successful communication between systems
3	Service configuration	Apache2	Web service running on Ubuntu
4	Service verification	Browser / curl	Apache web page accessible
5	Network scanning	Nmap	Ports and services identified
6	Service detection	Nmap -sV	Service and version information obtained
7	Traffic capture	Wireshark	Network packets captured
8	Protocol analysis	Wireshark filters	ICMP, HTTP, and DNS traffic identified

Overall Key Findings

The following key findings were obtained from the five tasks completed during Week 4:

1. A controlled cybersecurity laboratory environment was successfully established using Oracle VirtualBox with Kali Linux and Ubuntu Linux virtual machines.
2. The virtual machines were connected through a Host-only network, enabling controlled communication between the attacker and target systems.
3. The Apache2 web server was successfully installed and configured on the Ubuntu target machine.
4. The configured web service was accessed and verified from Kali Linux using the target machine's Host-only IP address.
5. Nmap was successfully used to identify accessible ports and services on the Ubuntu target and to perform service and version detection.
6. Wireshark was used to capture and analyze network traffic generated during the laboratory activities.
7. ICMP, HTTP, and DNS traffic were successfully identified using appropriate Wireshark display filters.
8. The activities demonstrated a complete basic cybersecurity workflow involving network configuration, service deployment, network scanning, and traffic analysis.
9. The laboratory environment provided a controlled platform for performing authorized cybersecurity testing without affecting external systems.

Overall Learning Outcomes

Upon completion of the five tasks, the following practical skills and concepts were gained:

- Understanding of virtual network configuration using Oracle VirtualBox.
- Practical experience in configuring Kali Linux and Ubuntu Linux within a controlled network.
- Understanding of Host-only networking and IP addressing.
- Verification of network connectivity using ICMP and the ping command.
- Installation, configuration, and verification of the Apache web server.
- Understanding of basic network services and their associated ports.
- Practical use of Nmap for network scanning and service identification.
- Understanding of service and version detection using Nmap.
- Practical experience in capturing network packets using Wireshark.
- Identification and analysis of ICMP, HTTP, and DNS protocols.
- Use of Wireshark display filters for protocol-specific traffic analysis.
- Understanding of the relationship between network services, exposed ports, and network traffic.
- Experience in documenting cybersecurity activities using command outputs and screenshots.
- Understanding of the importance of performing security testing only within an authorized environment.

Overall Conclusion

Week 4 provided practical experience in establishing and analyzing a controlled cybersecurity laboratory environment. The five tasks collectively covered the complete workflow from setting up a virtual network to configuring a web service, scanning the target system, and analyzing the resulting network traffic.

Kali Linux was used as the primary security testing and analysis machine, while Ubuntu Linux served as the target system. The Apache web server provided a practical network service for testing, Nmap enabled network and service enumeration, and Wireshark provided packet-level visibility into network communication.

The activities helped develop a practical understanding of network configuration, service management, network scanning, packet capture, and protocol analysis. Overall, the week strengthened the foundational skills required for further learning and practical work in networking and cybersecurity.